QUAN MINH TRAN, MD, PhD

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RESEARCH PROFILE

Infectious disease modeler with an MD and PhD in Biological Sciences, specializing in arboviral disease dynamics, vaccine impact modeling, serological analysis, Bayesian modeling, and outbreak analytics. Current CDC Prevention Effectiveness Fellow in the Dengue Branch, with first-author publications on yellow fever, chikungunya, and Japanese encephalitis modeling. Research program focuses on integrating surveillance, serological, and mechanistic modeling approaches to guide vaccination policy, outbreak response, and global disease burden estimation.

RESEARCH INTERESTS

Mathematical modeling of infectious diseases; arboviral disease dynamics; dengue epidemiology; vaccine impact modeling; seroepidemiology; Bayesian hierarchical modeling; epidemic nowcasting and forecasting; reproductive number estimation; global disease burden estimation; outbreak analytics; public health decision support.

EDUCATION

University of Notre Dame **Notre Dame, IN**
Ph.D., Biological Sciences | August 2023
Dissertation: Rethinking and improving models used to estimate the global disease burden of yellow fever
Advisor: Alex Perkins | GPA: 3.706

Pham Ngoc Thach University of Medicine **Ho Chi Minh City, Vietnam**
M.D., Doctor of Medicine | November 2016
GPA: 7.38

RESEARCH EXPERIENCE

Centers for Disease Control and Prevention **San Juan, Puerto Rico**
National Center for Emerging and Zoonotic Infectious Diseases, Division of Vector-Borne Diseases, Dengue Branch
Prevention Effectiveness Fellow | August 2023-present

- Developed catalytic modeling approaches to identify biases in dengue case reporting, inform vaccination strategies, and characterize serotype-specific transmission dynamics.
- Conducted nowcasting and forecasting analyses to support dengue outbreak preparedness and response.
- Estimated instantaneous reproductive numbers for dengue to characterize changes in transmission intensity.
- Developed a global dengue risk and burden map.
- Supported dengue outbreak response activities in Puerto Rico through analytic, surveillance, and modeling contributions.
- Mentored fellows and provided technical support to cross-functional teams within the Dengue Branch.

University of Notre Dame **Notre Dame, IN**
Department of Biological Sciences
Graduate Research Assistant | January 2019-August 2023

- Developed a modeling framework to identify eligible sites for future chikungunya vaccine trials, resulting in a first-author publication in Vaccine.
- Collaborated with UNICEF Brazil to implement forecasting methods for dengue fever in Brazil.
- Projected yellow fever burden and assessed vaccination impact for the Vaccine Impact Modeling Consortium.
- Modified and improved yellow fever burden projection models by reassessing age-pattern assumptions, quantifying uncertainty in vaccination coverage data, and incorporating human-to-human transmission.
- Provided modeling and epidemiologic input across multiple collaborative projects, resulting in several coauthored publications.

Oxford University Clinical Research Unit **Ho Chi Minh City, Vietnam**
Mathematical Modeling in Infectious Diseases Department
Research Assistant | August 2016-December 2018

- Analyzed microarray slide images for two research projects, contributing to two coauthored manuscripts.
- Reconstructed historical chikungunya virus activity using catalytic models, resulting in a first-author publication in PLOS Neglected Tropical Diseases.
- Assessed the impact of Japanese encephalitis vaccination programs using mathematical modeling, resulting in a first-author publication in eLife.

TECHNICAL SKILLS

Programming and workflows: R, Git/GitHub, Unix shell, reproducible analysis workflows, data visualization.

Modeling and statistical methods: catalytic models, Bayesian hierarchical modeling, epidemic nowcasting and forecasting, reproductive number estimation, vaccine impact modeling, disease burden estimation, cost effectiveness analysis, serological data analysis.
Disease areas: dengue, yellow fever, chikungunya, Japanese encephalitis, influenza, COVID-19, pertussis, Rift Valley fever.

SELECTED FIRST-AUTHOR PUBLICATIONS

Mapping global dengue transmission intensity, environmental suitability, and disease burden
QM Tran, MB Thayer, MA Johansson
In clearance

Lessons learned from real-time nowcasting: The 2024 dengue outbreak in Puerto Rico
QM Tran, A Detmar, CY Liu, ZJ Madewell, DM Rodriguez, JT Aponte, M Marzán-Rodriguez, G Paz-Bailey, LE Adams, K Holcomb, MA Johansson, MB Thayer
Submitted to Emerging Infectious Diseases. doi: 10.64898/2026.07.20.26358497

Misclassification of yellow fever vaccination status revealed through hierarchical Bayesian modeling
QM Tran, TA Perkins
American Journal of Epidemiology, March 2025. doi: 10.1093/aje/kwae465/8090263

Expected endpoints from future chikungunya vaccine trial sites informed by serological data and modeling
QM Tran, KJ Soda, AS Siraj, SM Moore, HE Clapham, TA Perkins
Vaccine, January 2023. doi: 10.1016/j.vaccine.2022.11.028

Estimating the global burden of Japanese encephalitis and the impact of vaccination: a systematic review combined with mathematical modelling
QM Tran, TTN Thao, NM Duy, TM Nhat, H Clapham
eLife, May 2020. doi: 10.7554/eLife.51027

Evidence of previous but not current transmission of chikungunya virus in southern and central Vietnam: Results from a systematic review and a seroprevalence study in four locations
QM Tran, HT Phuong, NHT Vy, NTL Thanh, NTN Lien, TTK Hong, PN Dung, NVV Chau, MF Boni, H Clapham
PLOS Neglected Tropical Diseases, February 2018. doi: 10.1371/journal.pntd.0006246

PEER-REVIEWED PUBLICATIONS AND MANUSCRIPTS

Why the growth of arboviral diseases necessitates a new generation of global risk maps and future projections
OJ Brady, LS Bastos, JM Caldwell, S Cauchemez, HE Clapham, I Dorigatti, KAM Gaythorpe, W Hu, L Hussain-Alkhateeb, MA Johansson, A Lim, V Lopez, RJ Maude, JP Messina, EA Mordecai, AT Peterson, I Rodriquez-Barraquer, IB Rabe, DP Rojas, SJ Ryan, H Salje, JC Semenza, QM Tran
PLOS Computational Biology, April 2025. doi: 10.1371/journal.pcbi.1012771

Misclassification of yellow fever vaccination status revealed through hierarchical Bayesian modeling
QM Tran, TA Perkins
American Journal of Epidemiology, March 2025. doi: 10.1093/aje/kwae465/8090263

Expected endpoints from future chikungunya vaccine trial sites informed by serological data and modeling
QM Tran, KJ Soda, AS Siraj, SM Moore, HE Clapham, TA Perkins
Vaccine, January 2023. doi: 10.1016/j.vaccine.2022.11.028

Ignoring transmission dynamics leads to underestimation of the impact of a novel intervention against mosquito-borne disease
SM Cavany, JH Huber, A Wieler, M Elliott, QM Tran, G Espana, SM Moore, TA Perkins
Manuscript submitted to Nature Medicine. doi: 10.1101/2021.11.19.21266602

Projecting vaccine demand and impact for emerging zoonotic pathogens
A Lerch, QA ten Bosch, M L'Azou-Jackson, AA Bettis, M Bernuzzi, GAV Murphy, QM Tran, JH Huber, A Siraj, M Elliott, C Hartlage, K Koh, K Strimbu, M Walters, TA Perkins, SM Moore
BMC Medicine, June 2022. doi: 10.1186/s12916-022-02405-1

Estimates of Japanese encephalitis mortality and morbidity: a systematic review and modeling analysis
Y Cheng, NM Tran, QM Tran, S Khandelwal, H Clapham
PLOS Neglected Tropical Diseases, May 2022. doi: 10.1371/journal.pntd.0010361

Age-seroprevalence curves for the multi-strain structure of influenza A virus
DN Vinh, NTD Nhat, E Bruin, NHT Vy, TTN Thao, HT Phuong, PH Anh, S Todd, QM Tran, NTL Thanh, NTN Lien, NTH Ha, TTK Hong, PQ Thai, M Choisy, TD Nguyen, CP Simmons, GE Thwaites, H Clapham, NVV Chau, M Koopmans, MF Boni
Nature Communications 12:6680, November 2021

Timing is everything when it comes to pertussis vaccination
TA Perkins, QM Tran
Lancet Infectious Diseases, October 2021. doi: 10.1016/S1473-3099(21)00353-4

Burden is in the eye of the beholder: sensitivity of yellow fever disease burden estimates to modeling assumptions
TA Perkins, JH Huber, QM Tran, RJ Oidtman, MK Walters, AS Siraj, SM Moore
Science Advances 7:eabg5033, October 2021

Impacts of K-12 school reopening on the COVID-19 epidemic in Indiana, USA
G Espana, SM Cavany, RJ Oidtman, C Barbera, A Costello, A Lerch, M Poterek, QM Tran, A Wieler, S Moore, TA Perkins
Epidemics 37:100487, August 2021

Lives saved with vaccination for 10 pathogens across 112 countries in a pre-COVID-19 world
J Toor, S Echeverria-Londono, X Li, K Abbas, ED Carter, HE Clapham, A Clark, MJ de Villiers, K Eilertson, M Ferrari, I Gamkrelidze, TB Hallett, WR Hinsley, D Hogan, JH Huber, ML Jackson, K Jean, A Karachaliou, P Klepac, A Kraay, J Lessler, BA Lopman, T Mengistu, CJE Metcalf, SM Moore, S Nayagam, T Papadopoulos, TA Perkins, A Portnoy, H Razavi, D Razavi-Shearer, S Resch, C Sanderson, S Sweet, Y Tam, H Tanvir, QM Tran, CL Trotter, SA Truelove, E Vynnycky, N Walker, A Winter, K Woodruff, NM Ferguson, KAM Gaythorpe
eLife 10:e67635, July 2021

Over 100 years of Rift Valley fever: a patchwork of data on pathogen spread and spillover
GM Bron, K Strimbu, H Cecilia, A Lerch, SM Moore, QM Tran, TA Perkins, QA ten Bosch
Pathogens 10:708, June 2021

Lying in wait: the resurgence of dengue virus after the Zika epidemic in Brazil
AF Brito, LC Machado, RJ Oidtman, MJL Siconelli, QM Tran, JR Fauver, RD Carvalho, FZ Dezordi, MR Pereira, LA deCastro-Jorge, ECM Minto, LMR Passos, CC Kalinich, ME Petrone, E Allen, G Espana, AT Huang, DAT Cummings, G Baele, RFO Franca, BAL da Fonseca, TA Perkins, GL Wallau, ND Grubaugh
Nature Communications 12:2619, May 2021

Estimating the health impact of vaccination against ten pathogens in 98 low-income and middle-income countries from 2000 to 2030: a modelling study
X Li, C Mukandavire, ZM Cucunuba, S Echeverria Londono, K Abbas, HE Clapham, M Jit, HL Johnson, T Papadopoulos, E Vynnycky, M Brisson, ED Carter, A Clark, MJ de Villiers, K Eilertson, MJ Ferrari, I Gamkrelidze, KAM Gaythorpe, NC Grassly, TB Hallett, W Hinsley, ML Jackson, K Jean, A Karachaliou, P Klepac, J Lessler, SM Moore, S Nayagam, DM Nguyen, H Razavi, D Razavi-Shearer, S Resch, C Sanderson, S Sweet, S Sy, Y Tam, H Tanvir, QM Tran, CL Trotter, S Truelove, K van Zandvoort, S Verguet, N Walker, A Winter, K Woodruff, NM Ferguson, T Garske
Lancet 397:398-408, January 2021

Estimating the global burden of Japanese encephalitis and the impact of vaccination: a systematic review combined with mathematical modelling
QM Tran, TTN Thao, NM Duy, TM Nhat, H Clapham
eLife, May 2020. doi: 10.7554/eLife.51027

Evidence of previous but not current transmission of chikungunya virus in southern and central Vietnam: Results from a systematic review and a seroprevalence study in four locations
QM Tran, HT Phuong, NHT Vy, NTL Thanh, NTN Lien, TTK Hong, PN Dung, NVV Chau, MF Boni, H Clapham
PLOS Neglected Tropical Diseases, February 2018. doi: 10.1371/journal.pntd.0006246

Structure of general-population antibody titer distributions to influenza A virus
NTD Nhat, S Todd, E Bruin, TTN Thao, NHT Vy, QM Tran, DN Vinh, J Beek, PH Anh, HM Lam, NT Hung, NTL Thanh, HLA Huy, VTH Ha, S Baker, GE Thwaites, NTN Lien, TTK Hong, J Farrar, CP Simmons, NVV Chau, M Koopmans, MF Boni
Scientific Reports 7:6060, July 2017

Women's Communication Self-Efficacy and Expectations of Primary Male Partners' Cooperation in Sexually Transmitted Infection Treatment in Ho Chi Minh City, Vietnam: A Cross-Sectional Study
LTH Tran, TC Bui, CM Markham, MD Swartz, QM Tran, AG Nyitray, TTT Huynh, LY Hwang
BMC Public Health, January 2016. doi: 10.1186/s12889-016-2690-0

Self-Reported Oral Health, Oral Hygiene, and Oral HPV Infection in At-Risk Women in Ho Chi Minh City, Vietnam
TC Bui, LTH Tran, CM Markham, TT Huynh, LT Tran, VTT Pham, QM Tran, NH Hoang, LY Hwang
Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology, July 2015. doi: 10.1016/j.oooo.2015.04.004

PRESENTATIONS

Integrating age-stratified case and serological data to inform dengue vaccination programs. Poster presentation, Dengue Endgame, August 2024.

Burden is in the eye of the beholder: sensitivity of yellow fever disease burden estimates to data types and modeling assumptions. Oral presentation, Ecological Society of America, August 2022.

Accounting for uncertainty in yellow fever vaccination status in estimates of vaccination coverage. Poster presentation, Ecology and Evolution of Infectious Diseases, May 2022.

Accounting for uncertainty in yellow fever vaccination status in estimates of vaccination coverage and disease burden. Poster presentation, 8th International Conference on Infectious Disease Dynamics, November 2021.

Expected endpoints from future chikungunya vaccine trial sites informed by serological data and modeling. Poster presentation, American Society of Tropical Medicine and Hygiene, November 2020.

Estimating sample size for future chikungunya vaccine trials based on projections of epidemic size that account for pre-existing immunity. Oral presentation, 7th International Conference on Infectious Disease Dynamics, December 2019.

Estimating the burden of Japanese encephalitis and the impact of vaccination. Oral presentation, Oxford Tropical Network Conference, March 2018.

Seroprevalence of chikungunya in Vietnam: evidence of past but not present transmission. Poster presentation, American Society of Tropical Medicine and Hygiene 66th Annual Meeting, November 2017.

TEACHING EXPERIENCE

Centers for Disease Control and Prevention **Atlanta, GA**
Lecturer, Introduction to R Programming | May 2025
- Developed curriculum, coding exercises, and practical materials for a four-day course introducing CDC employees to foundational R programming and GitHub.
- Delivered lectures and led interactive coding sessions for CDC participants.

University of Notre Dame **Notre Dame, IN**
Teaching Assistant, Department of Biological Sciences | January 2022-December 2022
- Introduction to Biocomputing: delivered weekly lectures, graded student exercises, and supported students in Unix shell, R programming, and GitHub.
- Infectious Disease Epidemiology: supported lectures, grading, and student learning in epidemiologic principles and quantitative approaches.

Infectious Disease Modeling Workshop **Chennai, India**
Teaching Assistant | November 2018
- Led a practical session and supported participants with code during a workshop introducing local policymakers to serological data analysis using mathematical modeling methods.

FELLOWSHIPS, GRANTS, HONORS, AND AWARDS

- Honor Award for outstanding leadership in public health response to the dengue and Oropouche virus outbreaks, 2025
- Eck Institute for Global Health Graduate Research Fellowship, 2022-2023
- Summer Institute in Statistics and Modeling in Infectious Diseases Scholarship, 2022
- EcoHealthNet Research Exchange Program, 2020
- Rapid Exposure to Advanced Computation Training Award (REACT), 2019
- Clinic on Meaningful Modeling of Epidemiological Data, ICI3D MMED Workshop Scholarship and Travel Award, 2019
- Travel Award, Computational Biology for Infectious Diseases, Quy Nhon, Vietnam, 2017
- Best Student Oral Presentation Award, conference presentation on electrophysiological modeling in generalized epilepsy, 2016

ACADEMIC SERVICE

Peer reviewer for 19 manuscripts: PLOS Neglected Tropical Diseases (8), PLOS ONE (1), PLOS Computational Biology (1), PLOS Medicine (1), Scientific Reports/Nature Scientific Reports (2), Nature Communications (1), Emerging Infectious Diseases (2), and BMC Public Health (3).

Grant reviewer: Modeling Infectious Diseases in Healthcare (MInD Healthcare) to Improve Pathogen Prevention and Healthcare Delivery; opportunity number CDC-RFA-CK-25-0018.

GOVERNMENT SERVICE

Joined outbreak response task force for dengue in Puerto Rico, 2024-2025

LEADERSHIP AND OUTREACH

Project mentor, Conference for Diversity in Mathematical Modeling and Public Health, January 2022. Coached undergraduate students in developing and presenting a research proposal related to mathematical modeling.

PROFESSIONAL AFFILIATIONS

Vaccine Impact Modeling Consortium, Member, 2017-2023. Contributed modeling analyses to assess the impact of Japanese encephalitis and yellow fever vaccination programs as part of a five-year consortium coordinated by Imperial College London.

REFERENCES

Available upon request.